

- (5 pt) 1. Briefly explain the difference between an assemblage and a community.
- (10 pt) 2. Explain the relationship between the Antarctic Convergence and the Tropical Submergence as an influence on the vertical distribution of deep-sea fishes in the Southern Ocean and more equatorial regions.
- (10 pt) 3. List two adaptations that Antarctic fishes without hemoglobin must have to take up and distribute oxygen throughout its body. Briefly and specifically explain how each adaptation allows the fish to survive without hemoglobin in the Antarctic region.

- (10 pt) 4. List two adaptations of bathypelagic fishes to low productivity. Briefly and specifically explain how each adaptation relates to low productivity and how they allow the fishes to survive in the bathypelagic.
- (10 pt) 5. List two sensory adaptations of cave fishes to the dark environment. Briefly and specifically explain how each adaptation relates to the absence of light and how they allow the fishes to survive in the cave environment.
- (10 pt) 6. **(GSH)** Both caves and the bathypelagic zone receive absolutely no light from the sun. However, cave fishes lose pigmentation and are uniformly pale but bathypelagic fishes produce pigment and are uniformly dark. Develop a hypothesis to explain why bathypelagic fishes might maintain black bodies in the total absence of sunlight. (I have a specific answer in mind but I am interested primarily in your reasoning and explanation.)

- (15 pt) 7. Name and describe three different mechanisms or “filters” that ultimately determine which species of fishes occur at a given locality. Provide a distinct example for each filter, and explain how each contributes to the observed assemblage.

- (20 pt) 8. Your friend has just returned from a diving trip to Indonesia. She asks how so many species of fishes can co-exist on a reef, and why do the species present seem to vary from reef to reef, and from time to time. Use your knowledge of competition, predation, larval availability and random events to develop a hypothesis that would explain your friend's observations. You may emphasize any combination of the four items listed above but you must discuss all four. This question is not about a "correct" answer but integration and explanation of the concepts. Write a hypothesis that is biologically sound and scientifically accurate, based on our class discussion.

- (20 pt) 9. Schlosser (1987) developed a predictive conceptual framework to describe changes in fish communities along a habitat complexity gradient in a stream. He argues that the relative importance of physical (environmental) versus biological processes changes along the stream gradient. For this question, consider two stream segments: a second order, high gradient section of an Ozark stream and a sixth order, low gradient section of the same stream. For each segment, compare and contrast the relative importance of physical and biological processes that affect assemblage structure. Explain each of the specific physical and biological processes (including habitat structure and complexity, and age classes of fishes) that Schlosser would argue are most important at each segment. Include in your explanation how these processes contribute to the stability or variability of the community at each segment. List 3–4 representative species typical of each community (use scientific names for maximum points). We also discussed these ideas in class. There's more room on the back if you need.